

# ***Decision Tree***

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*Decision tree classifier*

| Day | Outlook  | Temp | Humidity | Wind   | Play Tennis |
|-----|----------|------|----------|--------|-------------|
| D1  | Sunny    | Hot  | High     | Weak   | No          |
| D2  | Sunny    | Hot  | High     | Strong | No          |
| D3  | Overcast | Hot  | High     | Weak   | Yes         |
| D4  | Rain     | Mild | High     | Weak   | Yes         |
| D5  | Rain     | Cool | Normal   | Weak   | Yes         |
| D6  | Rain     | Cool | Normal   | Strong | No          |
| D7  | Overcast | Cool | Normal   | Strong | Yes         |
| D8  | Sunny    | Mild | High     | Weak   | No          |
| D9  | Sunny    | Cool | Normal   | Weak   | Yes         |
| D10 | Rain     | Mild | Normal   | Weak   | Yes         |
| D11 | Sunny    | Mild | Normal   | Strong | Yes         |
| D12 | Overcast | Mild | High     | Strong | Yes         |
| D13 | Overcast | Hot  | Normal   | Weak   | Yes         |
| D14 | Rain     | Mild | High     | Strong | No          |

## Attribute: Outlook

Values (Outlook) = Sunny, Overcast, Rain

$$S = [9+, 5-]$$

$$Entropy(S) = -\frac{9}{14} \log_2 \frac{9}{14} - \frac{5}{14} \log_2 \frac{5}{14} = 0.94$$

$$S_{Sunny} \leftarrow [2+, 3-]$$

$$Entropy(S_{Sunny}) = -\frac{2}{5} \log_2 \frac{2}{5} - \frac{3}{5} \log_2 \frac{3}{5} = 0.971$$

$$S_{Overcast} \leftarrow [4+, 0-]$$

$$Entropy(S_{Overcast}) = -\frac{4}{4} \log_2 \frac{4}{4} - \frac{0}{4} \log_2 \frac{0}{4} = 0$$

$$S_{Rain} \leftarrow [3+, 2-]$$

$$Entropy(S_{Rain}) = -\frac{3}{5} \log_2 \frac{3}{5} - \frac{2}{5} \log_2 \frac{2}{5} = 0.971$$

$$Gain(S, Outlook) = Entropy(S) - \sum_{v \in \{Sunny, Overcast, Rain\}} \frac{|S_v|}{|S|} Entropy(S_v)$$

$$Gain(S, Outlook)$$

$$= Entropy(S) - \frac{5}{14} Entropy(S_{Sunny}) - \frac{4}{14} Entropy(S_{Overcast}) - \frac{5}{14} Entropy(S_{Rain})$$

$$Gain(S, Outlook) = 0.94 - \frac{5}{14} 0.971 - \frac{4}{14} 0 - \frac{5}{14} 0.971 = 0.2464$$

| Day | Outlook  | Temp | Humidity | Wind   | Play Tennis |
|-----|----------|------|----------|--------|-------------|
| D1  | Sunny    | Hot  | High     | Weak   | No          |
| D2  | Sunny    | Hot  | High     | Strong | No          |
| D3  | Overcast | Hot  | High     | Weak   | Yes         |
| D4  | Rain     | Mild | High     | Weak   | Yes         |
| D5  | Rain     | Cool | Normal   | Weak   | Yes         |
| D6  | Rain     | Cool | Normal   | Strong | No          |
| D7  | Overcast | Cool | Normal   | Strong | Yes         |
| D8  | Sunny    | Mild | High     | Weak   | No          |
| D9  | Sunny    | Cool | Normal   | Weak   | Yes         |
| D10 | Rain     | Mild | Normal   | Weak   | Yes         |
| D11 | Sunny    | Mild | Normal   | Strong | Yes         |
| D12 | Overcast | Mild | High     | Strong | Yes         |
| D13 | Overcast | Hot  | Normal   | Weak   | Yes         |
| D14 | Rain     | Mild | High     | Strong | No          |

Attribute: Temp

Values (Temp) = Hot, Mild, Cool

$S = [9+, 5 -]$

$Entropy(S) = -\frac{9}{14}log_2\frac{9}{14} - \frac{5}{14}log_2\frac{5}{14} = 0.94$

$S_{Hot} \leftarrow [2+, 2-]$

$Entropy(S_{Hot}) = -\frac{2}{4}log_2\frac{2}{4} - \frac{2}{4}log_2\frac{2}{4} = 1.0$

$S_{Mild} \leftarrow [4+, 2-]$

$Entropy(S_{Mild}) = -\frac{4}{6}log_2\frac{4}{6} - \frac{2}{6}log_2\frac{2}{6} = 0.9183$

$S_{Cool} \leftarrow [3+, 1-]$

$Entropy(S_{Cool}) = -\frac{3}{4}log_2\frac{3}{4} - \frac{1}{4}log_2\frac{1}{4} = 0.8113$

$Gain(S, Temp) = Entropy(S) - \sum_{v \in \{Hot, Mild, Cool\}} \frac{|S_v|}{|S|} Entropy(S_v)$

$Gain(S, Temp)$

$= Entropy(S) - \frac{4}{14} Entropy(S_{Hot}) - \frac{6}{14} Entropy(S_{Mild})$

$- \frac{4}{14} Entropy(S_{Cool})$

$Gain(S, Temp) = 0.94 - \frac{4}{14} 1.0 - \frac{6}{14} 0.9183 - \frac{4}{14} 0.8113 = 0.0289$

| Day | Outlook  | Temp | Humidity | Wind   | Play Tennis |
|-----|----------|------|----------|--------|-------------|
| D1  | Sunny    | Hot  | High     | Weak   | No          |
| D2  | Sunny    | Hot  | High     | Strong | No          |
| D3  | Overcast | Hot  | High     | Weak   | Yes         |
| D4  | Rain     | Mild | High     | Weak   | Yes         |
| D5  | Rain     | Cool | Normal   | Weak   | Yes         |
| D6  | Rain     | Cool | Normal   | Strong | No          |
| D7  | Overcast | Cool | Normal   | Strong | Yes         |
| D8  | Sunny    | Mild | High     | Weak   | No          |
| D9  | Sunny    | Cool | Normal   | Weak   | Yes         |
| D10 | Rain     | Mild | Normal   | Weak   | Yes         |
| D11 | Sunny    | Mild | Normal   | Strong | Yes         |
| D12 | Overcast | Mild | High     | Strong | Yes         |
| D13 | Overcast | Hot  | Normal   | Weak   | Yes         |
| D14 | Rain     | Mild | High     | Strong | No          |

## Attribute: Humidity

Values (Humidity) = High, Normal

$$S = [9+, 5-]$$

$$Entropy(S) = -\frac{9}{14} \log_2 \frac{9}{14} - \frac{5}{14} \log_2 \frac{5}{14} = 0.94$$

$$S_{High} \leftarrow [3+, 4-]$$

$$Entropy(S_{High}) = -\frac{3}{7} \log_2 \frac{3}{7} - \frac{4}{7} \log_2 \frac{4}{7} = 0.9852$$

$$S_{Normal} \leftarrow [6+, 1-]$$

$$Entropy(S_{Normal}) = -\frac{6}{7} \log_2 \frac{6}{7} - \frac{1}{7} \log_2 \frac{1}{7} = 0.5916$$

$$Gain(S, Humidity) = Entropy(S) - \sum_{v \in \{High, Normal\}} \frac{|S_v|}{|S|} Entropy(S_v)$$

$$Gain(S, Humidity)$$

$$= Entropy(S) - \frac{7}{14} Entropy(S_{High}) - \frac{7}{14} Entropy(S_{Normal})$$

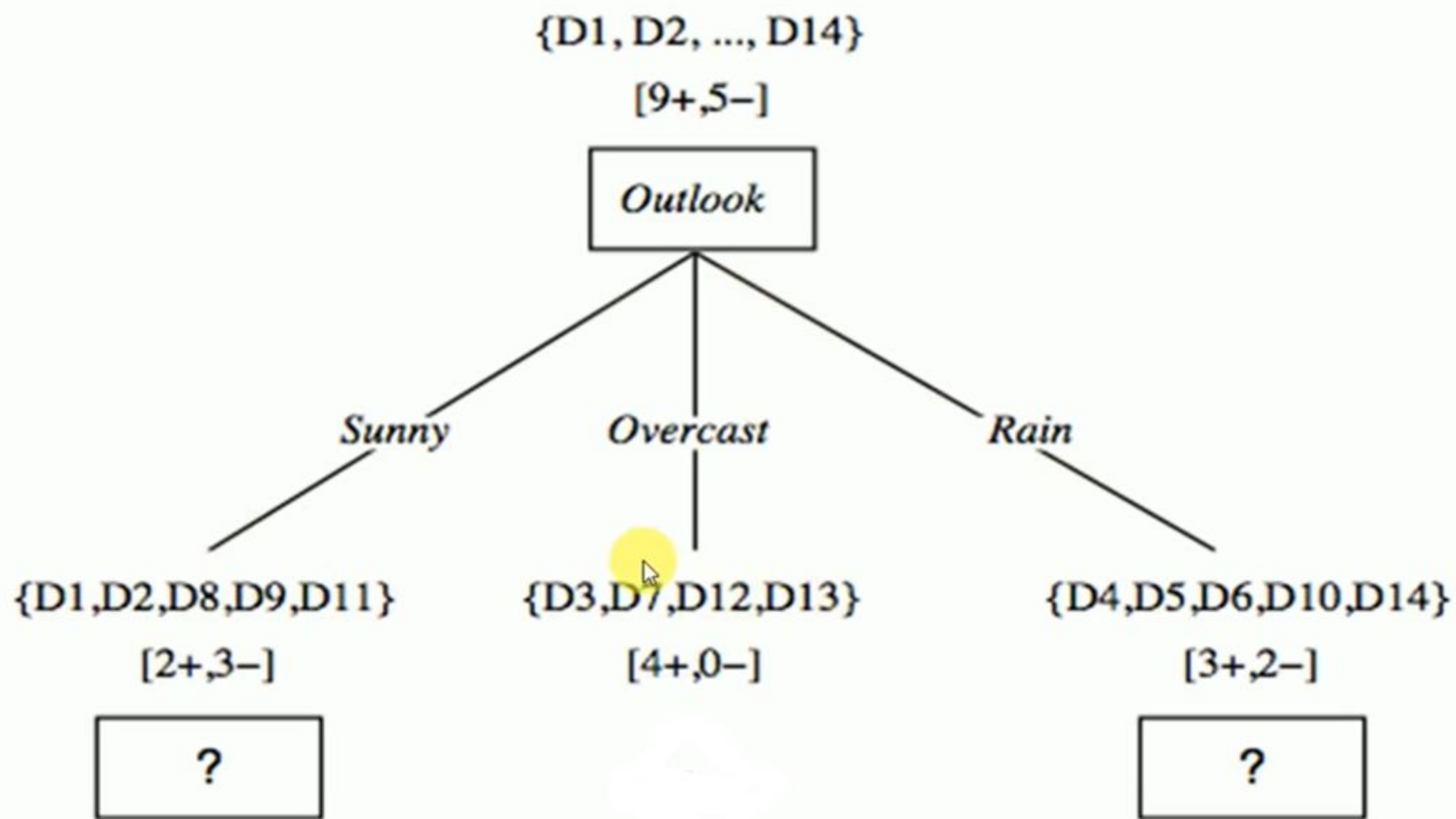
$$Gain(S, Humidity) = 0.94 - \frac{7}{14} 0.9852 - \frac{7}{14} 0.5916 = 0.1516$$

$$\textit{Gain}(S, \textit{Outlook}) = 0.2464$$

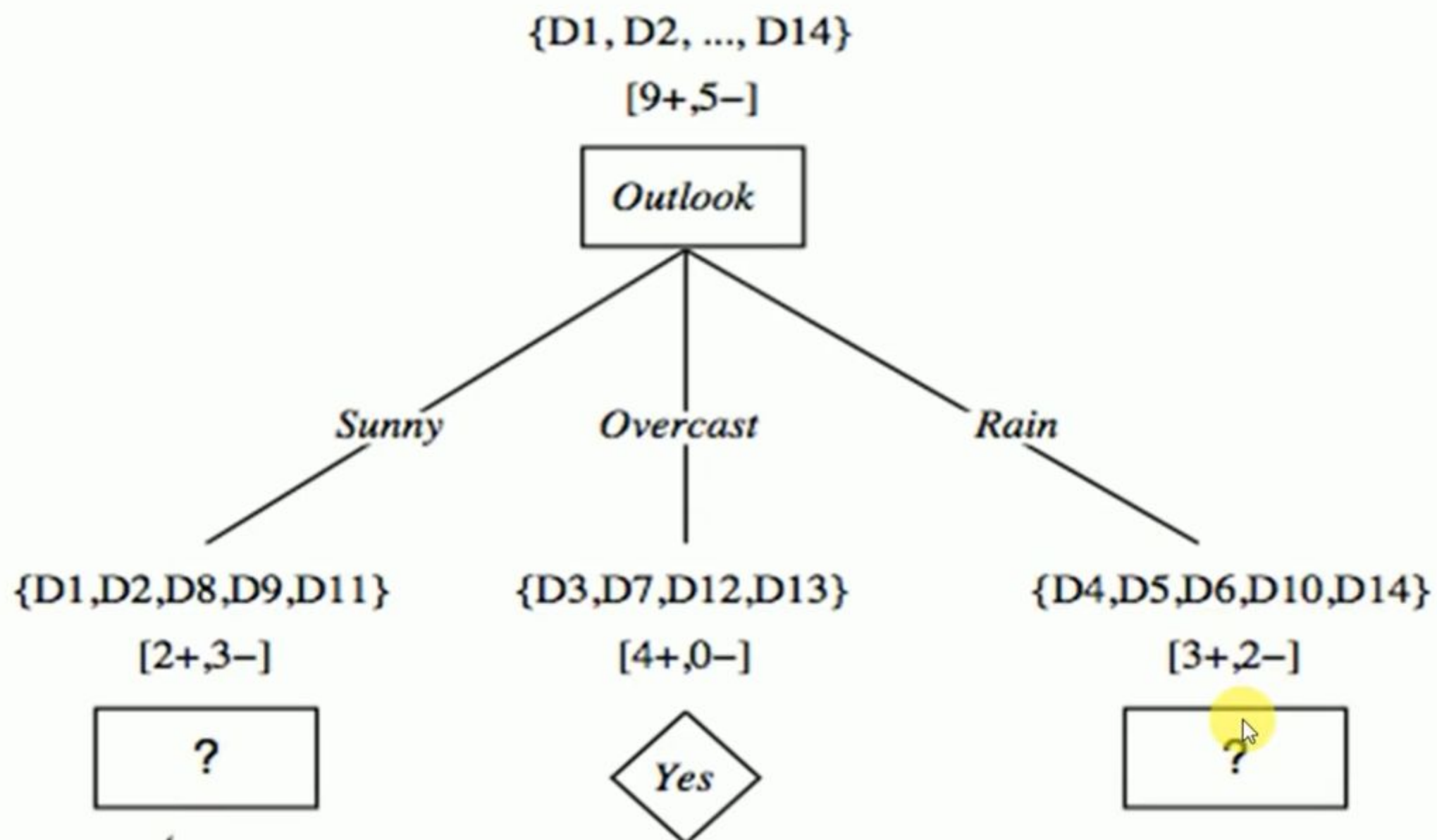
$$\textit{Gain}(S, \textit{Temp}) = 0.0289$$

$$\textit{Gain}(S, \textit{Humidity}) = 0.1516$$

$$\textit{Gain}(S, \textit{Wind}) = 0.0478$$



| Day | Outlook  | Temp | Humidity | Wind   | Play Tennis |
|-----|----------|------|----------|--------|-------------|
| D1  | Sunny    | Hot  | High     | Weak   | No          |
| D2  | Sunny    | Hot  | High     | Strong | No          |
| D3  | Overcast | Hot  | High     | Weak   | Yes         |
| D4  | Rain     | Mild | High     | Weak   | Yes         |
| D5  | Rain     | Cool | Normal   | Weak   | Yes         |
| D6  | Rain     | Cool | Normal   | Strong | No          |
| D7  | Overcast | Cool | Normal   | Strong | Yes         |
| D8  | Sunny    | Mild | High     | Weak   | No          |
| D9  | Sunny    | Cool | Normal   | Weak   | Yes         |
| D10 | Rain     | Mild | Normal   | Weak   | Yes         |
| D11 | Sunny    | Mild | Normal   | Strong | Yes         |
| D12 | Overcast | Mild | High     | Strong | Yes         |
| D13 | Overcast | Hot  | Normal   | Weak   | Yes         |
| D14 | Rain     | Mild | High     | Strong | No          |



| Day | Temp | Humidity | Wind   | Play Tennis |
|-----|------|----------|--------|-------------|
| D1  | Hot  | High     | Weak   | No          |
| D2  | Hot  | High     | Strong | No          |
| D8  | Mild | High     | Weak   | No          |
| D9  | Cool | Normal   | Weak   | Yes         |
| D11 | Mild | Normal   | Strong | Yes         |



| Day | Outlook  | Temp | Humidity | Wind   | Play Tennis |
|-----|----------|------|----------|--------|-------------|
| D1  | Sunny    | Hot  | High     | Weak   | No          |
| D2  | Sunny    | Hot  | High     | Strong | No          |
| D3  | Overcast | Hot  | High     | Weak   | Yes         |
| D4  | Rain     | Mild | High     | Weak   | Yes         |
| D5  | Rain     | Cool | Normal   | Weak   | Yes         |
| D6  | Rain     | Cool | Normal   | Strong | No          |
| D7  | Overcast | Cool | Normal   | Strong | Yes         |
| D8  | Sunny    | Mild | High     | Weak   | No          |
| D9  | Sunny    | Cool | Normal   | Weak   | Yes         |
| D10 | Rain     | Mild | Normal   | Weak   | Yes         |
| D11 | Sunny    | Mild | Normal   | Strong | Yes         |
| D12 | Overcast | Mild | High     | Strong | Yes         |
| D13 | Overcast | Hot  | Normal   | Weak   | Yes         |
| D14 | Rain     | Mild | High     | Strong | No          |

## Attribute: Temp

Values (Temp) = Hot, Mild, Cool

$$S_{Sunny} = [2+, 3-]$$

$$Entropy(S_{Sunny}) = -\frac{2}{5}\log_2\frac{2}{5} - \frac{3}{5}\log_2\frac{3}{5} = 0.97$$

$$S_{Hot} \leftarrow [0+, 2-]$$

$$Entropy(S_{Hot}) = 0.0$$

$$S_{Mild} \leftarrow [1+, 1-]$$

$$Entropy(S_{Mild}) = 1.0$$

$$S_{Cool} \leftarrow [1+, 0-]$$

$$Entropy(S_{Cool}) = 0.0$$

$$Gain(S_{Sunny}, Temp) = Entropy(S) - \sum_{v \in \{Hot, Mild, Cool\}} \frac{|S_v|}{|S|} Entropy(S_v)$$

$$Gain(S_{Sunny}, Temp)$$

$$= Entropy(S) - \frac{2}{5} Entropy(S_{Hot}) - \frac{2}{5} Entropy(S_{Mild})$$

$$- \frac{1}{5} Entropy(S_{Cool})$$

$$Gain(S_{sunny}, Temp) = 0.97 - \frac{2}{5} 0.0 - \frac{2}{5} 1 - \frac{1}{5} 0.0 = 0.570$$

| Day | Temp | Humidity | Wind   | Play Tennis |
|-----|------|----------|--------|-------------|
| D1  | Hot  | High     | Weak   | No          |
| D2  | Hot  | High     | Strong | No          |
| D8  | Mild | High     | Weak   | No          |
| D9  | Cool | Normal   | Weak   | Yes         |
| D11 | Mild | Normal   | Strong | Yes         |

## Attribute: Humidity

Values (Humidity) = High, Normal

$$S_{Sunny} = [2+, 3-]$$

$$Entropy(S) = -\frac{2}{5} \log_2 \frac{2}{5} - \frac{3}{5} \log_2 \frac{3}{5} = 0.97$$

$$S_{high} \leftarrow [0+, 3-]$$

$$Entropy(S_{High}) = 0.0$$

$$S_{Normal} \leftarrow [2+, 0-]$$

$$Entropy(S_{Normal}) = 0.0$$

$$Gain(S_{Sunny}, Humidity) = Entropy(S) - \sum_{v \in \{High, Normal\}} \frac{|S_v|}{|S|} Entropy(S_v)$$

$$Gain(S_{Sunny}, Humidity) = Entropy(S) - \frac{3}{5} Entropy(S_{High}) - \frac{2}{5} Entropy(S_{Normal})$$

$$Gain(S_{Sunny}, Humidity) = 0.97 - \frac{3}{5} 0.0 - \frac{2}{5} 0.0 = 0.97$$

| Day | Temp | Humidity | Wind   | Play Tennis |
|-----|------|----------|--------|-------------|
| D1  | Hot  | High     | Weak   | No          |
| D2  | Hot  | High     | Strong | No          |
| D8  | Mild | High     | Weak   | No          |
| D9  | Cool | Normal   | Weak   | Yes         |
| D11 | Mild | Normal   | Strong | Yes         |

## Attribute: Wind

Values (Wind) = Strong, Weak

$$S_{Sunny} = [2+, 3-]$$

$$Entropy(S) = -\frac{2}{5} \log_2 \frac{2}{5} - \frac{3}{5} \log_2 \frac{3}{5} = 0.97$$

$$S_{Strong} \leftarrow [1+, 1-]$$

$$Entropy(S_{Strong}) = 1.0$$

$$S_{Weak} \leftarrow [1+, 2-]$$

$$Entropy(S_{Weak}) = -\frac{1}{3} \log_2 \frac{1}{3} - \frac{2}{3} \log_2 \frac{2}{3} = 0.9183$$

$$Gain(S_{Sunny}, Wind) = Entropy(S) - \sum_{v \in \{Strong, Weak\}} \frac{|S_v|}{|S|} Entropy(S_v)$$

$$Gain(S_{Sunny}, Wind) = Entropy(S) - \frac{2}{5} Entropy(S_{Strong}) - \frac{3}{5} Entropy(S_{Weak})$$

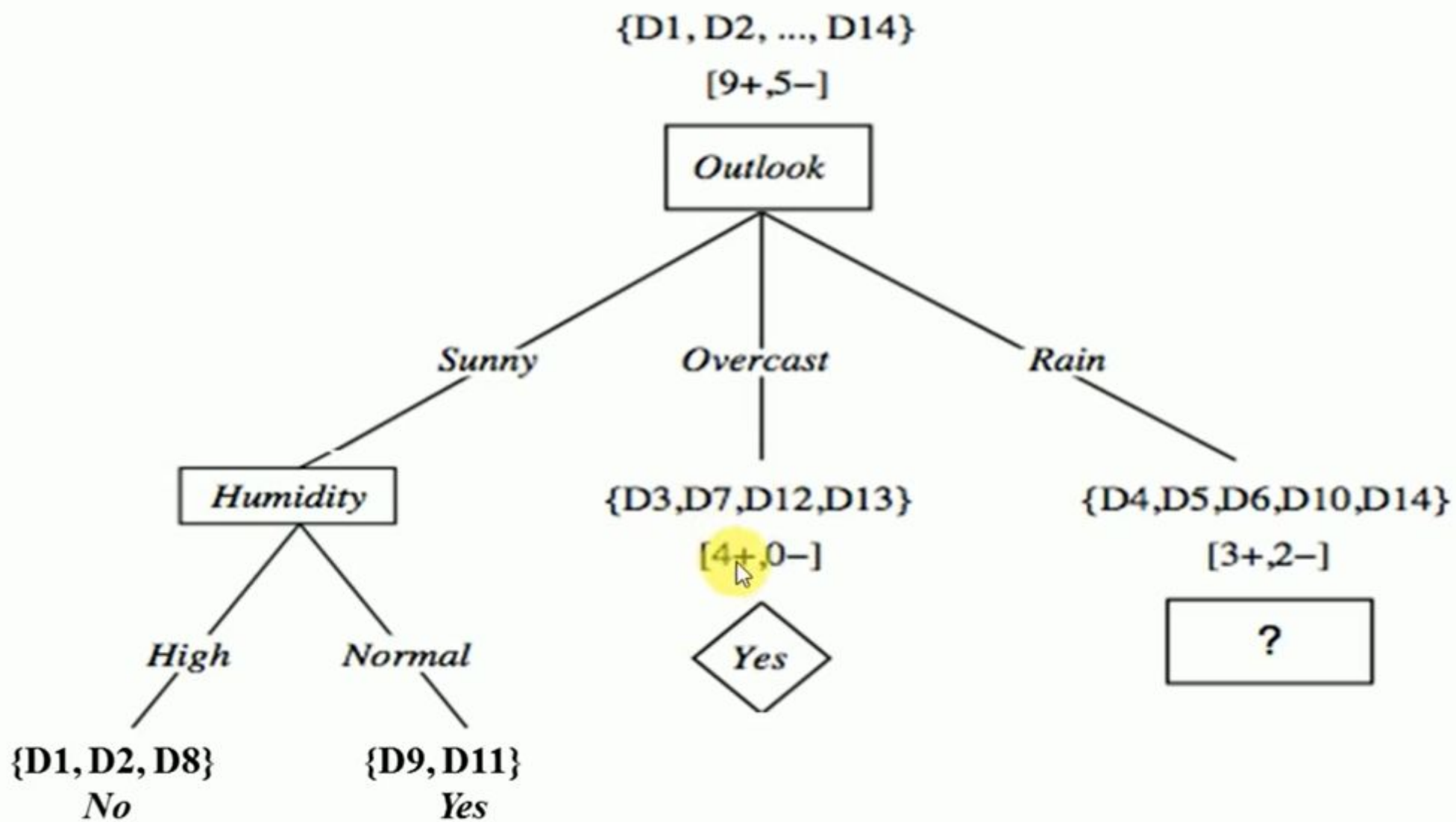
$$Gain(S_{sunny}, Wind) = 0.97 - \frac{2}{5} 1.0 - \frac{3}{5} 0.918 = 0.0192$$

| Day | Temp | Humidity | Wind   | Play Tennis |
|-----|------|----------|--------|-------------|
| D1  | Hot  | High     | Weak   | No          |
| D2  | Hot  | High     | Strong | No          |
| D8  | Mild | High     | Weak   | No          |
| D9  | Cool | Normal   | Weak   | Yes         |
| D11 | Mild | Normal   | Strong | Yes         |

$$\textit{Gain}(S_{\textit{sunny}}, \textit{Temp}) = 0.570$$

$$\textit{Gain}(S_{\textit{sunny}}, \textit{Humidity}) = 0.97$$

$$\textit{Gain}(S_{\textit{sunny}}, \textit{Wind}) = 0.0192$$



| Day | Temp | Humidity | Wind   | Play Tennis |
|-----|------|----------|--------|-------------|
| D4  | Mild | High     | Weak   | Yes         |
| D5  | Cool | Normal   | Weak   | Yes         |
| D6  | Cool | Normal   | Strong | No          |
| D10 | Mild | Normal   | Weak   | Yes         |
| D14 | Mild | High     | Strong | No          |



| Day | Outlook  | Temp | Humidity | Wind   | Play Tennis |
|-----|----------|------|----------|--------|-------------|
| D1  | Sunny    | Hot  | High     | Weak   | No          |
| D2  | Sunny    | Hot  | High     | Strong | No          |
| D3  | Overcast | Hot  | High     | Weak   | Yes         |
| D4  | Rain     | Mild | High     | Weak   | Yes         |
| D5  | Rain     | Cool | Normal   | Weak   | Yes         |
| D6  | Rain     | Cool | Normal   | Strong | No          |
| D7  | Overcast | Cool | Normal   | Strong | Yes         |
| D8  | Sunny    | Mild | High     | Weak   | No          |
| D9  | Sunny    | Cool | Normal   | Weak   | Yes         |
| D10 | Rain     | Mild | Normal   | Weak   | Yes         |
| D11 | Sunny    | Mild | Normal   | Strong | Yes         |
| D12 | Overcast | Mild | High     | Strong | Yes         |
| D13 | Overcast | Hot  | Normal   | Weak   | Yes         |
| D14 | Rain     | Mild | High     | Strong | No          |

## Attribute: Temp

Values (Temp) = Hot, Mild, Cool

$$S_{Rain} = [3+, 2-]$$

$$Entropy(S_{Sunny}) = -\frac{3}{5}\log_2\frac{3}{5} - \frac{2}{5}\log_2\frac{2}{5} = 0.97$$

$$S_{Hot} \leftarrow [0+, 0-]$$

$$Entropy(S_{Hot}) = 0.0$$

$$S_{Mild} \leftarrow [2+, 1-]$$

$$Entropy(S_{Mild}) = -\frac{2}{3}\log_2\frac{2}{3} - \frac{1}{3}\log_2\frac{1}{3} = 0.9183$$

$$S_{Cool} \leftarrow [1+, 1-]$$

$$Entropy(S_{Cool}) = 1.0$$

$$Gain(S_{Rain}, Temp) = Entropy(S) - \sum_{v \in \{Hot, Mild, Cool\}} \frac{|S_v|}{|S|} Entropy(S_v)$$

$$Gain(S_{Rain}, Temp)$$

$$= Entropy(S) - \frac{0}{5} Entropy(S_{Hot}) - \frac{3}{5} Entropy(S_{Mild}) - \frac{2}{5} Entropy(S_{Cool})$$

$$Gain(S_{Rain}, Temp) = 0.97 - \frac{0}{5} 0.0 - \frac{3}{5} 0.918 - \frac{2}{5} 1.0 = 0.0192$$

| Day | Temp | Humidity | Wind   | Play Tennis |
|-----|------|----------|--------|-------------|
| D4  | Mild | High     | Weak   | Yes         |
| D5  | Cool | Normal   | Weak   | Yes         |
| D6  | Cool | Normal   | Strong | No          |
| D10 | Mild | Normal   | Weak   | Yes         |
| D14 | Mild | High     | Strong | No          |

# Attribute: Humidity

Values (Humidity) = High, Normal

$$S_{Rain} = [3+, 2-]$$

$$Entropy(S_{Sunny}) = -\frac{3}{5}log_2\frac{3}{5} - \frac{2}{5}log_2\frac{2}{5} = 0.97$$

$$S_{High} \leftarrow [1+, 1-]$$

$$Entropy(S_{High}) = 1.0$$

$$S_{Normal} \leftarrow [2+, 1-]$$

$$Entropy(S_{Normal}) = -\frac{2}{3}log_2\frac{2}{3} - \frac{1}{3}log_2\frac{1}{3} = 0.9183$$

$$Gain(S_{Rain}, Humidity) = Entropy(S) - \sum_{v \in \{High, Normal\}} \frac{|S_v|}{|S|} Entropy(S_v)$$

$$Gain(S_{Rain}, Humidity) = Entropy(S) - \frac{2}{5}Entropy(S_{High}) - \frac{3}{5}Entropy(S_{Normal})$$

$$Gain(S_{Rain}, Humidity) = 0.97 - \frac{2}{5} 1.0 - \frac{3}{5} 0.918 = 0.0192$$

| Day | Temp | Humidity | Wind   | Play Tennis |
|-----|------|----------|--------|-------------|
| D4  | Mild | High     | Weak   | Yes         |
| D5  | Cool | Normal   | Weak   | Yes         |
| D6  | Cool | Normal   | Strong | No          |
| D10 | Mild | Normal   | Weak   | Yes         |
| D14 | Mild | High     | Strong | No          |

## Attribute: Wind

Values (wind) = Strong, Weak

$$S_{Rain} = [3+, 2-]$$

$$Entropy(S_{Sunny}) = -\frac{3}{5}\log_2\frac{3}{5} - \frac{2}{5}\log_2\frac{2}{5} = 0.97$$

$$S_{Strong} \leftarrow [0+, 2-]$$

$$Entropy(S_{Strong}) = 0.0$$

$$S_{Weak} \leftarrow [3+, 0-]$$

$$Entropy(S_{Weak}) = 0.0$$

$$Gain(S_{Rain}, Wind) = Entropy(S) - \sum_{v \in \{Strong, Weak\}} \frac{|S_v|}{|S|} Entropy(S_v)$$

$$Gain(S_{Rain}, Wind) = Entropy(S) - \frac{2}{5} Entropy(S_{Strong}) - \frac{3}{5} Entropy(S_{Weak})$$

$$Gain(S_{Rain}, Wind) = 0.97 - \frac{2}{5} 0.0 - \frac{3}{5} 0.0 = 0.97$$

| Day | Temp | Humidity | Wind   | Play Tennis |
|-----|------|----------|--------|-------------|
| D4  | Mild | High     | Weak   | Yes         |
| D5  | Cool | Normal   | Weak   | Yes         |
| D6  | Cool | Normal   | Strong | No          |
| D10 | Mild | Normal   | Weak   | Yes         |
| D14 | Mild | High     | Strong | No          |

$$Gain(S_{Rain}, Temp) = 0.0192$$

$$Gain(S_{Rain}, Humidity) = 0.0192$$

$$Gain(S_{Rain}, Wind) = 0.97$$

